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Longitudinally evaluated the relationship between body fat percentage and the risk for type 2 diabetes mellitus: Korean Genome and Epidemiology Study (KoGES)

Running title: body fat and Diabetes Mellitus

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ABSTRACT

Background: Body fat plays the significant role in maintaining glucose homeostasis. However, it is not fully identified how body fat percentage (BF%) has an impact on the development of type 2 diabetes mellitus (T2DM). Thus, this study was to evaluate the incidental risk for T2DM according to BF% level.

Methods: In a community based Korean cohort, 5,972 Korean adults were divided into quintile groups by BF% and followed-up for 10 years to monitor the development of T2DM. Cox proportional hazard model was used to evaluate the hazard ratios (HRs) for T2DM according to BF% quintile. Additionally, subgroup analysis was conducted by low and high level of BF% (cutoff: 25 in men and 35 in women) and body mass index (BMI).

Results: In adjusted model, compared to the BF% quintile 1 group, the risk for T2DM significantly increased over BF% of 22.8% in men and 32.9% in women (≥quintile 4). The level of BF% related to the increased risk for T2DM was lower in non-obese men (22.8%) than obese men (28.4%). In subgroup analysis, men with low BMI (<25) and high BF% (≥ 25) had the highest risk for T2DM than other subgroups [HRs: 1.83 (1.33 – 2.52)]. However, this association didn't show the statistical significance in women [HRs: 1.63 (0.98 – 2.72)].

Conclusion: The incidental risk for T2DM significantly increased over the specific level of BF%, which was lower in non-obese population than obese population. Gender difference was suggested in the incidental relationship between BF% and T2DM.

KEY WORDS: TYPE 2 diabetes mellitus, body fat and obesity

INTRODUNCTION

A substantial number of epidemiological works have heralded the upcoming diabetic pandemic, and its subsequent threat on public health. Asia-pacific region is also troubled by the increasing clinical issues related to type 2 diabetes mellitus (T2DM). Asian countries accounts for more than 60% of the world's diabetic population (1), and in particular, almost half of the global T2DM patient is diagnosed in Asia-pacific region (2). Moreover, it has been recognized that Asians have a strong ethnic and genetic predisposition for diabetes with the lower thresholds for the environmental risk factors (1), compared to Caucasians. Thus, it would be a significant clinical achievement to identify the simple and effective ways to early predict the development T2DM in Asians. Obesity is an established risk factor for DM and characterized by an excessive amount of body fat. Previous studies have focused on the role of body fat in the pathogenesis of T2DM, which has been postulated to mediate the causative relationship of obesity with T2DM (3). Although obesity has been traditionally classified by body mass index (BMI), it has been demonstrated that BMI is not a good indicator for body fat content (4,5). Thus, anthropometric measures such as waist circumference (WC) or the waist-to-hip ratio were used as surrogate markers of body fat distribution (6,7). However, because these measures are adopted as indicators reflecting the abdominal adiposity, it is still not enough to evaluate the amount of body fat content.

Recent studies have demonstrated that body fat percentage (BF%) was significantly associated with the cardiometabolic risk factors such as elevated blood pressure, hyperglycemia and dyslipidemia regardless of BMI and abdominal obesity (8-11). Additionally, lean (BMI<25kg/m^2) and obese (BMI≥30kg/m^2) individuals with elevated BF% were more likely to have prediabetes or T2DM than those with normal BF% (12). However, data were still insufficient to identify the incidental relationship between BF% and T2DM.

Thus, using data from a cohort of Korean Genome and Epidemiology Study (KoGES), we conducted this study to examine the risk for T2DM according to BF% level.

RESEARCH DESIGN AND METHODS

Study population.

All subjects were participants of the KoGES Ansan and Ansung Study, which is a population-based, epidemiology study of rural and urban community in South Korea. Detailed methods and study population of KoGES described in the previous study (13). The baseline survey of KoGES Ansan and Ansung study was completed in 2001-2002, and follow-up survey were conducted every two years. Initially, a total of 10,038 participants aged 40 to 69 participated in the study. A total of 5,108 participants were recruited by cluster-sampling method stratified by age, sex, and residential district in Ansung community and 5,012 subjects selected by random sampling method in Ansan. Of these 10,038 participants, 1,510 had baseline diabetes or missing values in history of diabetes, A1C, fasting and 2h-glucose level. 1,829 participants further excluded because they didn't check bioelectrical impedance analysis (BIA) or missing values. During a 10 years follow-up period, 727 were excluded because follow-up loss or incomplete follow-up data. Finally, 5,972 participants were enrolled in the present study. All subjects participated in the study voluntarily, and informed consent was obtained in all cases. Ethics approvals for the study protocol and analysis of the data were obtained from the institutional review board of Kangbuk Samsung Hospital.

Clinical, biochemical and body composition measurements

Study data included a medical history and sociodemographic information provided by a self-administered questionnaire, anthropometric measurements and laboratory biochemical measurements. All study participants were also asked to respond to a health-related behavior questionnaire, which included the topics of alcohol consumption, smoking and exercise. Physical activity divided two categories; regular exercise (≥90 minutes exercise per week, at least moderate intensity) or inactive group. The questions about alcohol intake included the frequency of alcohol consumption on a weekly basis and the typical amount that was consumed on a daily basis (g/day). Smoking status divided three categories; never, former and current smoker. Diabetes mellitus was defined as fasting serum glucose level of at least 126 mg/dL, or serum HbA1c level of at least 6.5%, or 2h-glucose level at least 200 mg/dL, or participant have ever been diagnosed with diabetes. (14) Hypertension was defined in the participants who have ever been diagnosed with hypertension, or participants with the measured blood pressure (BP) ≥140/90 mmHg at initial examinations. BP was measured both arm in sitting position after participants had been in a relaxed state for at least 10 min. There was a 5-min resting period between each measurement. The arithmetic mean value of the BP was used to define the systolic and diastolic BP. All participant's height and weight were also measured and body mass index (BMI) was calculated (kg/m^2).

After fasting overnight for 12 h, the plasma concentrations of glucose, total cholesterol (TC), triglyceride (TG) and high density lipoprotein-cholesterol (HDL-C) were measured enzymatically using a 747 Chemistry Analyzer (Hitachi, Tokyo, Japan). In the baseline and follow-up examinations, the study participants also underwent OGTT. The fasting plasma insulin concentrations were determined by a radioimmunoassay kit (Linco Research,St. Charles, MO). The HbA1C level was measured by high-performance liquid chromatography (VARIANT II; Bio-Rad Laboratories, Hercules, CA). Insulin resistance was calculated by homeostasis model assessment-insulin resistance (HOMA-IR) and obtained using the

following formula: HOMA-IR= fasting serum insulin (uU/ mL) × fasting serum glucose (mg/dl)/405.

BF%, muscle mass and lean body mass were measured by tetrapolar bioelectrical impedance analysis (BIA) (Inbody 3.0, Biospace, Korea). Using formulas previously validated and empirically derived (15), BIA measures two parameters of fat and lean tissue and calculates BF%. BMI was calculated as the weight (kg) divided by the square of the height (m), and WC was measured in the standing position, at the level of umbilicus.

BMI ≥25 kg/m^2 was used as a cut-off of obesity based on the Asian-Pacific specific cutoff point of obesity (16), and BF% ≥25% in men and ≥35% in women were adopted as a cut-off of BF% based on the global cut-off of BF% defined in WHO report (17). Additionally, cutoff of WC was defined by 90 cm in men and 85 cm in women based on the criteria of the Korean Society of the Study of Obesity (18).

Statistical analysis

All study participants were divided into 5 groups according to their quintiles of BF% in each gender. Data are presented as means ± standard deviation for continuous variables and as proportions for categorical variables. To identify the linear trends of variables across the BF% quintile, we used linear regression model for continuous variables and Cochran-Armitage trend test for categorical variable.

The unadjusted and multivariate-adjusted hazard ratios (HR) and their 95% confidence intervals (C.I.) were estimated with the use of the Cox proportional hazards model. Each gender group was categorized into five groups according to their quintile of BF%. The covariates of the adjusted model are age, study area (Ansan or Ansung), hypertension, regular

exercise, smoking, alcohol intake, HOMA-IR, hsCRP, HDL-C and TG. The incidence cases of T2DM and incidence density (incidence cases per 1,000 person-years) was also calculated in each study groups. This analysis was also separately performed in each non-obese subjects (BMI <25 kg/m^2) and obese subjects (BMI ≥25 kg/m^2).

To evaluate the interaction of BMI and BF% on the risk for T2DM, subgroup analysis was conducted in 4 subgroups stratified by BMI cut-off and BF% cut-off. The cutoff of BMI was set in 25 kg/m^2 (low BMI <25 kg/m^2 and high BMI ≥25 kg/m^2), and BF% subgroup was divided by cutoff of 25% in men (low BF <25 % and high BF ≥25%) and 35% in women (low BF <35 % and high BF ≥35%). Subgroups consisted of 4 groups of low BMI with low BF%, low BMI with high BF%, high BMI with low BF% and high BMI with high BF%. Additionally, we conducted the subgroup analysis in each lowered BF% cut-off and WC cut-off instead BMI cut-off. BF% cut-off was lowered from 25% in male and 35% in female into 22.8% in men and 32.9% in women.

All statistical analyses were performed using R 3.3.3 (R Foundation for Statistical Computing, Vienna, Austria).

RESULT

Baseline clinical characteristics of the study population

A total of 5,972 participants were enrolled (2,831 men and 3,141 women). The mean age was 50.5 ± 8.4 in men and 51.3 ± 8.8 in women. The final follow-up examination was done in 2011-2012. During 10 years of follow-up, the development of T2DM was identified in 1,148 (584 men and 564 women). The overall incidence is 24.8 cases per 1,000 person-years. The baseline clinical characteristics of study population are presented in Table 1. In men, the baseline metabolic profiles like fasting glucose, HbA1c, TC, TG, HOMA-IR, BMI and WC

and the proportion of HTN tended to increase proportionally to the quintile of BF%. Quintile 1 group had the lowest muscle mass than other quintile groups. These findings were similarly observed in women.

Body Fat, BMI and the risk of T2DM

Table 2 showed the incidence case, incidence density, unadjusted and multivariate adjusted HRs for T2DM in men and women. The HRs for T2DM increased proportionally to the quintiles of BF%. When the quintile 1 of BF% was set as a reference, multivariate adjusted HRs for T2DM significantly increased above BF% level in quintile 4 in men (BF% ≥ 22.8%) and women (BF% ≥ 32.9%) [men: 1.81 (1.33-2.46) in quintile 4 and 2.14 (1.57 – 2.90) in quintile 5; women: 1.44 (1.07 – 1.93) in quintile 4 and 1.47 (1.09 – 1.98) in quintile 5].

Subgroup analyses by BMI level were presented in table 3 and 4. In non-obese group (BMI<25 kg/m^2), adjusted HRs for T2DM significantly increased from BF% level over 22.8% in quintile 5 [2.13 (1.45 – 3.13)]. In obese group (BMI≥25 kg/m^2), HRs for T2DM also significantly increased from quintile 5, which had the higher BF% cut-off (28.4%) than that in non-obese group.

Table 5 demonstrated the combined interaction of BMI with BF% on the risk for T2DM. In men, compared to group 1 (low BMI and low BF%), group 2 (low BMI and high BF) had the highest adjusted HRs for T2DM (1.83 [95% CI 1.38-2.52]). However, in women, group 2 (low BMI and high BF) lost the statistical significance in adjusted HRs for T2DM [1.63 (0.98 – 2.72)], and groups with high BMI (group 3 and 4) had the significantly increased adjusted HRs for T2DM, regardless of BF% [1.39 (1.12 – 1.73) in group 4 and 1.48 (1.20 – 1.82) in group 5]. The lean body mass (kg) was highest in group 3 (55.5 ± 4.7 in men and 40.4 ± 3.7 in women) and lowest in group 2(44.4 ± 4.3 in men and 36.3 ± 3.7 in women).

This finding was identically observed in the combined interaction of BMI with adjusted BF% from 25% in men and 35% in women to 22.8% in men and 32.9% in women (Supplementary table 1). However, when WC substituted BMI in analyzing the interaction of BMI with BF%, women showed the statistically significant association of T2DM with group 2 (low WC with high BF%) and group 4 (high WC and high BF%) (Supplementary table 2).

Discussion

It is widely believed that Asians, compared to Caucasians, had the higher cardiometabolic risk even at the given BMI. This sense is connected to a theory that Asian is more predisposed to T2DM than other ethnic groups. Actually, the prevalence of obesity is lower in Asians than Caucasians (19), whereas the prevalence of T2DM in Asia catches up with that of Western area (1,2). Additionally, T2DM patients with normal (BMI: 18-24.9) or low body weight (BMI<18) are more prevalent in Asia than Western countries (20). As a possible explanation for these facts, it has been discussed that the pathogenesis of T2DM is more likely to be influenced by body fat than obesity itself (21,22). We noticed the association between BF% and T2DM, and conducted this study on the basis of a hypothesis that BF% has the significant impact on the incidental risk for T2DM in Asians independently of BMI.

In the present study, men with BF% more than 22.8% (≥quintile 4) and women with BF% more than 32.9% (≥quintile 4) had the significantly increased risk for T2DM, compared to quintile 1 group. This finding indicates that BF% more than specific cutoff is clinically important as a risk factor for T2DM. Additionally, it may be suggested that previously addressed BF% cutoff of 25% in men and 35% in women is adjusted to the lower level in terms of predicting T2DM in Asians. Our observations are supported by previous studies displaying the association of BF% with adverse metabolic conditions closely linked to the

pathogenesis of T2DM. Longitudinally observational study for 6,171 US adults showed that normal weight obesity, defined as the combination of normal BMI and high body fat content, was associated with a high prevalence of metabolic dysregulation including metabolic syndrome (9). Additionally, several cross-sectional studies have suggested that elevation of BF% may increase the risk of T2DM. In a study for 4,828 white subjects aged 18-80 years, Gómez-Ambrosi et al found that BMI-classified lean individuals with prediabetes or T2DM had the higher BF% than BMI-classified lean individuals with normal glucose tolerance (12). A cross-sectional analysis for Korean showed that high BF% was significantly associated with cardiometabolic risk factors including hyperglycemia even in individuals within normal BMI categories (10). In particular, studies for Asians presented that BF% is more closely linked to the risk for T2DM in Asians than Caucasians. There is strong evidence that the metabolic consequences of obesity like insulin resistance and T2DM manifest at lower absolute amounts of total body fat in South Asians than in whites (23). In randomized controls for Japanese Americans with impaired glucose, treatment with aerobic exercise and life style modification including regulating animal fat and carbohydrate significantly lowered percent body fat, body weight and intra-abdominal fat at 6 and 24 months (24,25). In particular, at 24 months, prevalence of normal glucose tolerance was higher in the treatment group (40%) than control group (30%). These studies suggest that Asians are more susceptible to body fat in the pathogenesis of T2DM. As an explanation for these findings, we may raise the physical characteristics of Asians in body fat distribution. In abdominal CT, healthy Chinese and South Asians individuals were found to have a greater amount of visceral adipose tissue than Europeans with the same BMI or waist circumference (26). Additionally, multiethnic studies have highlighted that Asians have greater adiposity or visceral fat than Caucasians at any given BMI or waist circumference (27,28). It has been demonstrated that visceral adiposity measured by imaging technology is strongly associated

with the risk of diabetes in all ethnic groups, independent of BMI or WC (29-31). Thus, it is inferred that the increased risk for T2DM was attributable to the higher visceral adiposity at given body fat in Asians than Caucasians.

Our results suggest that BF% is more likely to contribute to the development of T2DM in non-obese individuals than obese individuals. The BF% linked to risk for T2DM was lower in non-obese men than obese men, and interaction test indicated that group 2 (low BMI and high BF%) had the highest risk for T2DM than other subgroups in men. The mechanism for these findings may be explained by results of lean body mass and WC in interaction test. In subgroup analysis, group 2 (low BMI and high BF%) had the lowest lean body mass and relatively higher WC than group 1 (low BMI and low BF%). Thus, it is postulated that the relatively lower muscle mass and higher abdominal obesity in group 2, despite of low BMI, had an impact on the increased risk of T2DM. These findings suggest that BF% may be an available clinical tool to predict the development of T2DM in non-obese men. Current guideline recommending the periodic screening for T2DM targets asymptomatic population with obesity or prediabetes like impaired glucose tolerance and impaired fasting glucose (32), but elevated BF% is not included in conditions needing screening for T2DM. However, our results suggest that increase of BF% per se has the predictability for T2DM as a potential risk factor. BF% can be checked by a noninvasive, simple and economical BIA. Thus, the finding that BF% predicts the future development of T2DM in non-obese men directs to a clinical implication that BF% is a potentially available clinical tool in T2DM surveillance for non-obese men easy to be overlooked for DM screening.

Our findings imply the gender difference in the influence of BF% on the pathogenesis of T2DM. While BF% level above quintile 5 (22.85% in non-obese men and 28.4% in obese men) was significantly associated with the increased risk for T2DM in both non-obese and obese men, women didn't show the statistically significant association between specific BF%

quintile and the risk for T2DM. Additionally, subgroup analysis for interaction test of BF% with BMI (WC) indicated the highest risk for T2DM in men with low BMI (WC) and high BF%, whereas women showed the diverse findings. These findings may be accounted for by the gender difference in BF%, amount of skeletal muscle, body fat distribution and hormonal secretion. Previous studies have reported the gender difference in the biological risk factors, health behavior and pathophysiological mechanisms of T2DM (33). Thus, further studies should be performed to identify the gender difference in the influence of body fat on the pathogenesis of T2DM.

The merit of our study is medical data obtained from a cohort of general Korean population with substantial identifiable medical records, which enable us to longitudinally evaluate the risk for DM according to the level of BF% and quantified the combined interaction of BF% and BMI on the incidental risk for T2DM. Nonetheless, several limitations and shortcomings should be recognized.

First, the limitations of bioelectrical impedance analysis may have impact on the results. It was reported that bioelectrical impedance analysis equations was insufficiently validated in obese individuals with BMI greater than 34 kg/m^2 (27). Morbidly obese individuals tend to have a relatively high amount of extracellular water and total body water, which may overestimate fat-free mass and underestimate fat mass (27). However, the mean BMI of our study subjects was 24.2 ± 2.8 in men and 24.7 ± 3.2 in women, which suggests that there was a scarce number of morbidly obese individuals in our study subjects. Thus, it isn't likely that our results were largely affected by the limitation of bioelectrical impedance analysis in morbidly obese individuals. Second, because our study was conducted only for a cohort of Korean, our findings should not be generalized in other ethnic group.

In conclusion, our study demonstrated that the increase of BF% was significantly associated

with the increased risk for T2DM. In Koreans, the risk for T2DM significantly increased at the level of BF% lower than the conventional cut-off of BF%, and the level of BF% linked to the increased risk for T2DM was lower in non-obese men than obese men. Additionally, men with low BMI and high BF% had the highest risk for T2DM in the interaction test of BMI and BF%, which was diversely demonstrated in women. These findings indicate the clinical significance of BF% in predicting T2DM, suggesting the gender difference in the influence of BF% on the T2DM.

Conflict of interest

Nothing to be declared. The authors report no relationships that could be construed as a conflict of interest.

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AUTHOR CONTRIBUTION

Sung Keun Park coordinated the study and wrote the manuscript as a first author. Chang-Mo Oh and Ju Young Jung played role in analyzing data and verifying the results. Jae-Hong Ryoo participated in conducting study and writing manuscript. Joong-Myung Choi participated in

reviewing manuscript.

Ju Young Jung is the guarantor of this work and, as such, had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Table 1. Baseline clinical characteristics of male and female study participants stratified by quintile of BF%

Characteristics	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5	P value
- Men (BF%)	(7.1 - 17.2)	(17.3 - 20.1)	(20.2 - 22.7)	(22.8 – 25.6)	(25.7 – 38.3)	
Age (year)	50.4 ± 8.6	50.1 ± 8.1	50.1 ± 8.5	50.1 ± 7.9	51.7 ± 8.7	0.029
Fasting Glucose (mg/dL)	82.4 ± 8.8	83.8 ± 8.6	84.8 ± 9.0	86.3 ± 9.1	86.0 ± 8.9	<0.001
HbA1c (%)	5.5 ± 0.3	5.5 ± 0.3	5.5 ± 0.3	5.6 ± 0.4	5.6 ± 0.3	<0.001
HOMA-IR	1.1 ± 0.6	1.4 ± 1.0	1.5 ± 0.8	1.6 ± 1.3	1.8 ± 1.0	<0.001
TC (mg/dL)	179.9 ± 33.6	187.3 ± 32.7	193.4 ± 34.7	198.6 ± 36.0	201.3 ± 32.8	<0.001
TG (mg/dL)	124.2 ± 79.7	160.3 ± 110.6	171.8 ± 100.0	194.9 ± 126.3	197.5 ± 104.1	<0.001
HDL-C (mg/dL)	48.4 ± 11.0	44.1 ± 9.6	43.3 ± 9.5	41.3 ± 8.5	41.2 ± 8.5	<0.001
BMI (kg/m^2)	21.2 ± 2.1	23.1 ± 1.8	24.4 ± 1.9	25.5 ± 1.9	27.0 ± 2.3	<0.001
WC (cm)	76.4 ± 5.6	81.4 ± 5.4	84.4 ± 5.3	87.4 ± 5.4	91.2 ± 5.6	<0.001
BF%	14.7 ± 2.0	18.8 ± 0.8	21.5 ± 0.7	24.1 ± 0.9	28.4 ± 2.0	<0.001

Muscle mass (kg)	48.2 ± 6.3	49.8 ± 5.8	49.8 ± 5.8	51.2 ± 6.0	50.1 ± 5.9	<0.001
Lean body mass (kg)	51.1 ± 6.6	52.7 ± 6.1	53.8 ± 6.2	54.1 ± 6.2	53.0 ± 6.2	<0.001
hsCRP	0.2 ± 0.8	0.2 ± 0.4	0.2 ± 0.3	0.2 ± 0.3	0.3 ± 0.4	0.144
Average alcohol use (g/day)	17.7 ± 29.2	17.4 ± 27.2	17.4 ± 27.2	16.6 ± 24.5	17.9 ± 29.7	0.883
Current smoking (%)	60.4%	48.8%	45.0%	41.8%	38.5%	<0.001
Regular exercise (%)	54.1%	39.8%	34.5%	31.2%	30.8%	<0.001
Hypertension (%)	18.6%	25.4%	27.2%	34.9%	41.1%	<0.001
- Women	(11.6 – 27.1)	(27.2 – 30.6)	(30.7 - 32.8)	(32.9 – 35.7)	(35.8 – 47.2)	
Age (year)	49.8 ± 8.9	49.8 ± 8.3	49.8 ± 8.3	51.5 ± 8.4	54.5 ± 8.7	<0.001
Fasting Glucose (mg/dL)	79.7 ± 7.2	80.6 ± 7.2	80.8 ± 7.4	81.5 ± 7.9	82.2 ± 8.0	<0.001
HbA1c (%)	5.4 ± 0.3	5.5 ± 0.3	5.5 ± 0.3	5.6 ± 0.4	5.6 ± 0.3	<0.001
HOMA-IR	1.3 ± 0.9	1.5 ± 1.0	1.6 ± 1.2	1.7 ± 0.9	1.8 ± 1.3	<0.001
TC (mg/dL)	178.2 ± 31.2	185.0 ± 32.1	188.8 ± 34.7	193.1 ± 32.7	200.2 ± 36.2	<0.001

TG (mg/dL)	112.4 ± 52.7	133.3 ± 77.2	144.0 ± 73.6	145.5 ± 70.8	163.5 ± 91.6	<0.001
HDL-C (mg/dL)	48.9 ± 10.3	46.8 ± 10.4	45.4 ± 9.9	45.2 ± 9.7	44.6 ± 9.5	<0.001
BMI (kg/m^2)	21.4 ± 2.0	21.4 ± 2.0	24.5 ± 1.8	25.7 ± 1.9	28.6 ± 2.7	<0.001
WC (cm)	73.8 ± 6.9	77.7 ± 7.3	81.3 ± 7.6	83.8 ± 7.7	90.0 ± 8.6	<0.001
BF%	23.8 ± 2.7	29.1 ± 1.0	31.8 ± 0.6	34.1 ± 0.8	38.5 ± 2.3	<0.001
Muscle mass (kg)	37.1 ± 4.4	37.4 ± 4.0	37.7 ± 4.1	37.7 ± 4.1	38.4 ± 4.6	<0.001
Lean body mass (kg)	39.4 ± 4.5	39.7 ± 4.2	40.0 ± 4.4	40.0 ± 4.3	40.8 ± 4.7	<0.001
hsCRP	0.1 ± 0.3	0.2 ± 0.3	0.2 ± 0.3	0.3 ± 0.5	0.3 ± 0.4	<0.001
Average alcohol use (g/day)	1.8 ± 8.9	1.0 ± 3.6	1.6 ± 5.9	1.4 ± 5.5	1.3 ± 5.0	0.376
Current smoking (%)	4.3%	4.3%	2.8%	3.1%	2.6%	0.029
Regular exercise (%)	39.3%	31.9%	34.7%	28.5%	28.5%	<0.001
Hypertension (%)	15.7%	19.3%	23.1%	28.5%	40.6%	<0.001

Continuous variables are expressed as mean (±SD), and categorical variables are expressed as number (percentage (%)).

TC: Total cholesterol, TG: triglyceride, HDL-C: high density lipoprotein cholesterol, BMI: body mass index, HOMA-IR: homeostasis model assessment of insulin resistance, BF%: body fat %

Table 2. Hazard ratios (HRs) and 95% confidence intervals (CI) for the incidence of type 2 diabetes mellitus according to BF% quintile

Characteristics	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5
- Men (BF%)	(7.1 - 17.2)	(17.3 - 20.1)	(20.2 - 22.7)	(22.8 – 25.6)	(25.7 – 38.3)
Unadjusted HR	1.00 (Reference)	1.38 (1.01 – 1.89)	1.56 (1.15 – 2.13)	2.40 (1.80 – 3.20)	3.09 (2.33 – 4.08)
Adjusted HR	1.00 (Reference)	1.16 (0.83 – 1.60)	1.27 (0.92 – 1.76)	1.81 (1.33 – 2.46)	2.14 (1.57 – 2.90)
Incidence cases	68	90	97	147	182
Incidence density	15.1	20.4	22.6	34.7	43.9
- Women (BF%)	(11.6 – 27.1)	(27.2 – 30.6)	(30.7 - 32.8)	(32.9 – 35.7)	(35.8 – 47.2)
Unadjusted HR	1.00 (Reference)	1.40 (1.04 – 1.88)	1.41 (1.05 – 1.89)	1.98 (1.49 – 2.62)	2.32 (1.76– 3.05)
Adjusted HR	1.00 (Reference)	1.15 (0.85 – 1.56)	1.10 (0.81 – 1.49)	1.44 (1.07 – 1.93)	1.47 (1.09 – 1.98)
Incidence cases	77	105	101	132	149
Incidence density	15.1	20.4	20.5	27.6	31.8

Adjusted for age, study area (Ansan or Ansung), hypertension, regular exercise, smoking, alcohol intake, HOMA-IR, hsCRP, HDL-C, TG

*Incidence density: incidence cases per 1,000 person-year

Table 3. Hazard ratios (HRs) and 95% confidence intervals (CI) for the incidence of type 2 diabetes mellitus according to BF% quintile in non-obese subjects (BMI<25 kg/m^2)

Characteristics	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5
- Men	(7.1 – 15.8)	(15.9 – 18.1)	(18.2 – 20.1)	(20.2 – 22.7)	(22.8 – 38.3)
Unadjusted HR	1.00 (Reference)	0.97 (0.63 – 1.49)	1.46 (0.99–2.16)	1.69 (1.15–2.49)	2.74 (1.92–3.91)
Adjusted HR	1.00 (Reference)	0.94 (0.61 – 1.45)	1.27 (0.84– 1.91)	1.47 (0.98–2.20)	2.13 (1.45–3.13)
Incidence cases	44	40	59	63	99
Incidence density	15.7	15.1	22.0	24.9	40.9
- Women	(11.6 – 24.9)	(25.0 – 27.9)	(28.0 -30.3)	(30.4 – 32.3)	(32.4 – 41.2)
Unadjusted HR	1.00 (Reference)	1.42 (0.94 – 2.15)	1.71 (1.14–2.56)	1.51 (1.001 – 2.27)	1.85 (1.23–2.79)
Adjusted HR	1.00 (Reference)	1.17 (0.76 – 1.81)	1.25 (0.82–1.91)	0.99 (0.64–1.53)	1.13 (0.73–1.74)
Incidence cases	39	52	58	56	57
Incidence density	13.1	17.9	19.9	19.0	21.4

Adjusted for age, study area (Ansan or Ansung), hypertension, regular exercise, smoking, alcohol intake, HOMA-IR, hsCRP, HDL-C, TG

*Incidence density: incidence cases per 1,000 person-year

Table 4. Hazard ratios (HRs) and 95% confidence intervals (CI) for the incidence of type 2 diabetes mellitus according to quintiles of body BF% in obese subjects (BMI≥25 kg/m^2)

Characteristics	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5
- Men	(12.2 – 21.9)	(22.0 – 23.7)	(23.8 – 25.8)	(25.9 – 28.3)	(28.4 – 35.9)
Unadjusted HR	1.00 (Reference)	1.38 (0.88 – 2.17)	1.98 (1.30–3.01)	2.02 (1.33–3.06)	2.81 (1.88– 4.20)
Adjusted HR	1.00 (Reference)	1.16 (0.73 – 1.84)	1.67 (1.08–2.57)	1.50 (0.97– 2.34)	1.94 (1.26– 2.98)
Incidence cases	34	43	61	62	79
Incidence density	18.8	25.3	35.4	36.4	50.1
- Women	(23.9 – 32.2)	(32.3 – 34.1)	(34.2 – 36.0)	(36.1 – 38.3)	(38.4 – 47.2)
Unadjusted HR	1.00 (Reference)	0.94 (0.65 – 1.37)	1.18 (0.82– 1.69)	1.32 (0.93–1.86)	1.17 (0.81–1.69)
Adjusted HR	1.00 (Reference)	0.85 (0.58 – 1.25)	1.14 (0.78– 1.65)	1.05 (0.73– 1.50)	0.94 (0.64– 1.38)
Incidence cases	58	54	61	71	58
Incidence density	27.4	25.4	30.8	34.1	30.1

Adjusted for age, study area (Ansan or Ansung), hypertension, regular exercise, smoking, alcohol intake, HOMA-IR, hsCRP, HDL-C, TG

*Incidence density: incidence cases per 1,000 person-year

Table 5. Hazard ratios (HRs) and 95% confidence intervals (CI) for the incidence of type 2 diabetes mellitus according to subgroups stratified by BMI cut-off in 25 kg/m^2 and BF% cut-off of 25% in men and 35% in women.

Characteristics	Group 1	Group 2	Group 3	Group 4
- Men				
Unadjusted HR	1.00 (Reference)	2.18 (1.59 – 2.98)	1.13 (0.90 – 1.41)	2.03 (1.67 – 2.46)
Adjusted HR	1.00 (Reference)	1.83 (1.33 – 2.52)	1.03 (0.82 – 1.30)	1.53 (1.24 – 1.89)
Incidence cases	259	46	109	170
Incidence density	21.5	44.7	24.2	42.3
Muscle mass (kg)	47.9 ± 5.2	44.4 ± 4.3	55.5 ± 4.7	51.8 ± 5.1
Lean body mass (kg)	50.7 ± 5.5	47.0 ± 4.5	58.7 ± 4.9	54.7 ± 5.4
BF%	18.5 ± 3.5	27.0 ± 2.0	22.0 ± 2.2	28.0 ± 2.2
BMI (kg/m^2)	22.3 ± 1.8	23.8 ± 1.0	26.4 ± 1.1	26.4 ± 1.1
Waist circumference (cm)	79.6 ± 5.8	85.8 ± 4.0	88.5 ± 4.7	92.0 ± 5.1
- Women				

Unadjusted HR	1.00 (Reference)	1.77 (1.07 – 2.94)	1.51 (1.23 – 1.87)	1.94 (1.60 – 2.36)
Adjusted HR	1.00 (Reference)	1.63 (0.98 – 2.72)	1.39 (1.12 – 1.73)	1.48 (1.20 – 1.82)
Incidence cases	246	16	134	168
Incidence density	17.7	30.0	26.5	32.5
Muscle mass (kg)	36.3 ± 3.7	32.6 ± 2.7	40.4 ± 3.7	38.9 ± 4.2
Lean body mass (kg)	38.6 ± 3.9	34.6 ± 2.9	42.8 ± 4.0	41.3 ± 4.4
BF%	28.3 ± 4.1	36.3 ± 1.3	32.2 ± 2.0	38.1 ± 2.5
BMI (kg/m^2)	22.5 ± 1.7	23.9 ± 1.0	26.4 ± 1.2	28.7 ± 2.4
Waist circumference (cm)	76.4 ± 7.3	80.9 ± 6.8	84.9 ± 7.0	90.1 ± 8.4

Adjusted for age, study area (Ansan or Ansung), hypertension, regular exercise, smoking, alcohol intake, HOMA-IR, hsCRP, HDL-C, TG

*Incidence density: incidence cases per 1,000 person-year

Group 1: low BMI (<25 kg/m^2) + low BF % (<25 in men, <35 in women)

Group 2: low BMI (<25 kg/m^2) + high BF % (≥25 in men, ≥35 in women)

Group 3: high BMI (≥25 kg/m^2) + low BF % (<25 in men, <35 in women)

Group 4: high BMI (≥25 kg/m^2) + high BF % (≥25 in men, ≥35 in women)